

ALEJANDRO MAYORAL

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Biomedical Engineer with experience in test engineering, manufacturing quality, incoming quality control (IQC), fixture design, and medical device development. Skilled in developing test protocols, supporting manufacturing operations, improving production processes, and implementing quality controls to increase reliability and reduce variability.

RELEVANT EXPERIENCE

Bioana — Product Development Engineering Intern

Monterrey, Mexico | Dec 2024 - Dec 2025

- Developed **test protocols, test reports, acceptance criteria**, and **work instructions** to support manufacturing quality and operator execution.
- **Designed and fabricated test and manufacturing fixtures** to support verification activities, improve repeatability, and reduce process variability.
- **Designed and implemented incoming quality control (IQC) tests** to evaluate components, materials, and manufacturing readiness before integration into production.
- **Executed experiments** and process evaluations in a **cleanroom manufacturing environment**, reducing process time by approximately **50%** in an intraocular lens production step.
- Participated in **failure analysis, PFMEA activities, and risk mitigation** initiatives to improve manufacturing robustness.
- **Investigated suppliers, materials, and production equipment** to support manufacturing and quality objectives.
- Collaborated with engineering and operations teams to **improve manufacturing processes** and define additional verification activities.

Research Assistant - Characterization of the coupling response between the BCI system-user through the analysis of EEG signals

Jan 2025 - June 2025

- Contributed to the project by conducting literature review, modifying code, processing data and analyzing results to identify patterns in EEG signals.
- Develop skills in signal processing, programming in MATLAB and Python, as well as interpretation of neurophysiological data.

EDUCATION AND COURSES

Universidad de Monterrey

Monterrey, Mexico | Aug 2021 - May 2026

B.S. in biomedical engineering - GPA: 3.9/4

Universidad Miguel Hernández

Elche, Spain | Jan 2026 - May 2026

Academic Exchange - Rehabilitation Technologies Research Project

- Designed and validated a rehabilitation assessment interface integrating a robotic system and EMG acquisition.
- Conducted structured testing and performance evaluation with users.
- Processed and analyzed test data using MATLAB and Python.
- Documented procedures, methodologies, and technical findings.

COURSES AND CERTIFICATIONS

- Course in medical devices: "Bachelor Master Classes por Boston Scientific".
- Course in Animal and Human Cell Culture: Applications in Regenerative Medicine (Universidad Complutense de Madrid).
- Certification as Microsoft Office Specialist: Excel
- Course in Data Analysis with Python by IBM

SKILLS

- Test Protocol & Report Development
- Incoming Quality Control (IQC)
- Root Cause Analysis
- PFMEA
- Fixture Design
- MATLAB, Python, Inventor, Fusion 360
- Microsoft Office: excel, word, power point
- Data analysis